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Highlights

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- A mathematical model for multi-region heterogeneous robot scheduling is established.
- A hybrid initialization strategy integrating MRNEH and IG heuristics is designed.
- A critical robot-oriented neighborhood search strategy is proposed.
- A round-based load balancing mechanism is developed to improve resource utilization.
- A scout bee phase with multi-operator searches and hybrid restart is introduced.

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An effective discrete artificial bee colony algorithm for multi-region heterogeneous picking robot collaborative scheduling

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Abstract

In smart farms, multiple types of picking robots are being progressively deployed across different regions, and scheduling these robots collaboratively has become a widespread challenge affecting picking efficiency. This paper investigates the multi-region heterogeneous picking robot task allocation and scheduling problem (MRHPRTAS). The problem is characterized by regional operation permission constraints, maximum capacity constraints, and heterogeneous picking robot capabilities, with the objective of minimizing the maximum completion time (Makespan). To address the multi-constraint coupling characteristics of this NP-hard problem, a multi-region heterogeneous discrete artificial bee colony (MRHDABC) algorithm is proposed, integrating a critical robot-oriented optimization strategy and a round-based load balancing strategy. MRHDABC employs multi-region NEH-based and IG-based heuristics to construct high-quality initial solutions, and achieves an effective balance between global exploration and local exploitation through a three-phase search mechanism. Extensive experiments are conducted on test instances covering diverse combinations of task scales and robot configurations, and the results show that MRHDABC consistently outperforms existing state-of-the-art algorithms across all problem scales, validating the effectiveness and robustness of the proposed algorithm.

Keywords: multi-region heterogeneous picking robot collaboration, task allocation and scheduling, discrete artificial bee colony algorithm, critical robot, load balancing

1. Introduction

With the rapid advancement of intelligent agriculture, agricultural robots have demonstrated increasing potential in improving productivity and reducing costs [1, 2].

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As a core application scenario of intelligent agriculture, smart farms are significantly transforming traditional agricultural production methods through the integration of various advanced technologies [3, 4]. In smart farms, picking is an important and complex task involving collaborative operations of heterogeneous picking robots across multiple regions [5, 6]. Efficiently allocating and scheduling these picking robots to complete tasks has become a key issue affecting system efficiency [7–9].

Picking tasks are distributed across different regions of smart farms and need to be executed by heterogeneous picking robots, which have different regional operation permissions and picking capability, and are subject to maximum capacity constraints [10]. In practice, after completing the last task of the current round, picking robots must return to the depot for unloading before starting the next round of tasks. These characteristics prevent simple task allocation strategies from obtaining high-quality scheduling solutions [11, 12]. In this context, it is necessary to determine the allocation of tasks to robots and the execution sequence of tasks for each robot, so as to minimize the maximum completion time (Makespan) among all robots.

The multi-region heterogeneous picking robot task allocation and scheduling problem (MRHPRTAS) is an NP-hard problem [13], posing significant challenges for obtaining optimal solution through exact algorithms within limited time. In recent years, researchers have proposed various metaheuristic algorithms to address similar scheduling problems, including ant colony optimization (ACO) [14], particle swarm optimization (PSO) [15], and artificial bee colony (ABC) algorithms [16, 17]. Among them, the discrete artificial bee colony (DABC) algorithm demonstrates significant advantages in solving complex scheduling problems due to its excellent balance between global exploration and local exploitation capabilities, as well as its relatively simple parameter configuration [18]. In practical smart farm operations, regional operation permissions restrict the regions in which each type of picking robot can execute tasks, maximum capacity constraints require robots to return to the depot for unloading after completing each round and before starting the next round of tasks, and robot heterogeneity is reflected in the differences in picking capability among robot types. These three characteristics are coupled with one another and jointly determine the feasibility and efficiency of task allocation and scheduling. However, existing DABC-based studies on agricultural robot task allocation, such as Chen et al. [16], Guo et al. [5], and Dai et al. [7], have yet to simultaneously address this coupling within a unified framework, which constitutes the core challenge of the MRHPRTAS investigated in this paper.

To address the above challenges, this paper proposes a multi-region heterogeneous discrete artificial bee colony (MRHDABC) algorithm. The algorithm integrates three innovative core strategies: multi-region heuristic initialization, optimization based on critical robot identification, and round-based load balancing. The main contributions of this paper can be summarized as follows:

1) A unified mathematical model integrating regional operation permissions, maximum capacity constraints, and heterogeneous characteristics is established, providing a complete modeling framework for multi-region heterogeneous robot scheduling problems.

2) A hybrid initialization strategy integrating multi-region NEH heuristic and IG mechanism is designed, improving the convergence efficiency and initial solution quality of the algorithm.

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3) A strategy based on critical robot identification is proposed, enhancing the algorithm's search capability in constrained spaces through the three-phase search mechanism.

The remainder of this paper is organized as follows: Section 2 reviews related research work; Section 3 describes the problem in detail and establishes the mathematical model; Section 4 introduces the proposed MRHDABC algorithm; Section 5 analyzes algorithm performance through experimental results; Section 6 summarizes the work and indicates future research directions.

2. Related works

The Multi-Robot Task Allocation (MRTA) problem represents an important research direction in robotics, with significant application value in agriculture. Chen et al. [19] tackle the cooperative task allocation problem for multiple UAVs in plant protection, employing Q-learning to determine the number of cooperating UAVs and thereby maximizing the total spraying area within a specified time limit. Targeting the cross-regional collaborative scheduling problem for agricultural machinery under time window constraints, Guo et al. [20] reduce the transfer distance and nonoperational scheduling cost through a population initialization strategy based on the time window priority rule. Wang et al. [21] investigate the coordinated scheduling problem of heterogeneous harvesters and grain trucks in harvesting-transportation scenarios, shortening the fleet's total operational time through a greedy heuristic initialization combined with a penalty-function-guided fitness evaluation mechanism. To address the task interruption problem caused by capacity constraints in agricultural harvesting scenarios, Guo et al. [22] construct a hybrid loading mode that supports task splitting, and balance the total energy consumption and the Makespan through a collaborative search strategy among parallel populations. Zhang et al. [23] tackle the task allocation and scheduling problem of multiple weeding robots in smart farms by designing a dual-heuristic initialization strategy combined with neighborhood search operators. Pan et al. [24] integrate the picking robot task scheduling problem with the vehicle dispatch scheduling problem, proposing a two-stage evolutionary algorithm incorporating a restart mechanism to simultaneously optimize the completion time and the delivery penalty. These studies provide an important methodological foundation for spatially distributed task allocation. However, when addressing the collaborative operation of multi-region heterogeneous picking robots, these studies have not adequately considered the multi-constraint coupling effects of regional operating permission constraints, maximum capacity constraints, and robot heterogeneity, making these approaches difficult to directly apply to the MRHPRTAS.

Heuristic and metaheuristic algorithms have become mainstream methods for solving MRTA problems due to their ability to obtain near-optimal solutions within acceptable computational time. In terms of agricultural task allocation, Guo et al. [25] address the sequential waiting constraint between harvesting and transportation vehicles by designing a hybrid crossover strategy based on greedy reinsertion. Targeting large-scale agricultural robot task allocation problems, Li et al. [26] design a topology-aware decomposition strategy that overcomes the spatial-task association characteristics that traditional geometric clustering methods struggle to accurately represent. Wang

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et al. [10] address the cooperative task allocation problem of weeding robots and spraying drones, simultaneously optimizing the Makespan and the total residual herbicide through heuristic initialization, a multi-teacher strategy, and a multi-neighborhood search mechanism. Dong et al. [11] set the total travel distance and the Makespan as optimization objectives, designing a crossover strategy guided by the non-dominated solution set along with a restart mechanism based on the ideal point. Chen et al. [27] address the multi-trip task scheduling problem of picking robots in smart orchards, optimizing the robots' total energy consumption under the Makespan constraint through an initialization strategy weighted by load and distance, along with a clustering-based local search mechanism.

In broader MRTA research, Huang et al. [28] address the cooperative task allocation problem for multiple spraying UAVs, achieving task balance and preventing flight conflicts through a dynamic block allocation strategy combined with a collision avoidance mechanism. Cao et al. [29] target the agricultural machinery overload and idleness problems caused by the "nearest allocation" strategy, designing an adaptive pheromone update strategy to accommodate new task arrivals and machinery failures. Kang et al. [30] improve the solution quality and local search capability for the multi-weeding robot task allocation problem through an enhanced heuristic initialization combined with a multi-neighborhood search strategy. Miao et al. [31] achieve a balanced optimization between the Makespan and the operating cost in weeding robot operation scheduling by designing a Q-learning-driven adaptive destruction strategy along with a local search mechanism. Wang et al. [32] enable the cooperative task allocation of pollination robots and spraying UAVs across multiple orchards through a cross-orchard balancing mechanism. In other multi-resource collaborative scheduling domains, Yang et al. [33] address the cost optimization challenge posed by stochastic economic factors in the joint scheduling of port berths, tugboats, and quay cranes, simultaneously minimizing the vessel's port time cost and the corresponding tugboat and crane operating costs through a mechanism integrating chaotic mapping and quantum rotation gates. Li et al. [34] formulate the berth-quay crane-experiment allocation problem at cargo and scientific research ports as a two-dimensional bin-packing problem with time windows, in which each vessel corresponds to a rectangle whose length equals the vessel's length and width equals its turnaround time, and reduce the average vessel turnaround time and the average variation of experiment start time through an improved genetic algorithm that schedules each vessel to the earliest feasible position.

Despite the rich theoretical and methodological foundations provided by existing research two deficiencies remain in addressing the MRHPRTAS.

1) Existing research mainly focuses on single constraints or simple heterogeneity and lacks both systematic modeling of the multi-constraint coupling effects of regional operation permissions, maximum capacity constraints, and robot heterogeneity, and specialized allocation strategies for the complex coupling problems addressed in this paper.

2) Existing algorithms often face challenges of local optima and insufficient solution diversity when handling large-scale multi-constraint scheduling problems, particularly in scenarios where picking robots need to execute tasks over multiple rounds.

To address these research gaps, this paper proposes the MRHDABC algorithm. The algorithm establishes a unified mathematical model that captures the multi-constraint

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coupling among regional operation permissions, maximum capacity constraints, and robot heterogeneity, and incorporates a multi-region heuristic initialization strategy, thereby providing a complete modeling foundation and high-quality initial solutions for task allocation in complex constrained environments. Furthermore, the algorithm employs a critical robot-oriented three-phase search mechanism to enhance its global search capability and solution diversity in large-scale scenarios, effectively solving the MRHPRTAS and improving the overall operational efficiency of the system.

3. Problem Description and Modeling

3.1. Problem description

Fig. 1 shows a schematic diagram of the multi-region task point distribution in a smart farm. The smart farm contains multiple task points distributed across different regions, with each region growing different types of crops. The coordinates and picking weights of task points requiring operation can be identified and calculated through sensor technology, and picking robots are dispatched to perform the picking tasks. The depot serves as both the starting and ending point for all picking robots.

To ensure efficient task execution, different types of heterogeneous picking robots need to be allocated to various regions. Although all picking robots can travel through any region, they can only execute picking operations within regions where they have regional operation permissions. To prevent collisions, picking robots depart from the depot in a fixed sequence with predetermined time intervals to execute picking operations. After completing each round of picking, the robots must return to the depot for unloading before starting the next round of tasks.

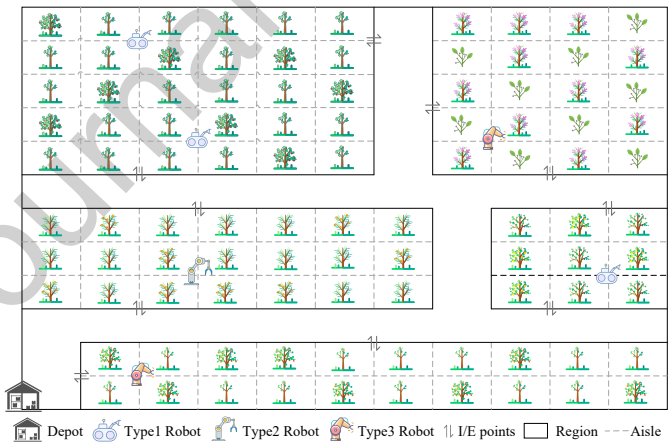


Fig. 1. Multi-regional task point distribution diagram.

The MRHPRTAS investigated in this paper can be described as follows: there are multiple task points distributed across different regions of the smart farm, and these tasks need to be assigned to different types of heterogeneous picking robots. The task assignment must consider the regional operation permission constraints and picking capability differences among different robot types. According to the actual situation,

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the number of picking robots is significantly fewer than the number of task points, i.e., $m \ll n$. The number of picking robots is pre-configured according to the task scale, with the optimization objective of determining the optimal task sequence for each robot to minimize the Makespan. Furthermore, without loss of generality, the following assumptions are made:

(1) All picking robots operate normally during task execution, without downtime, malfunctions, or other anomalies.

(2) All picking robots have identical maximum load capacities, and upon completing the last task of the current round, a picking robot must return to the depot for unloading.

(3) All tasks must be assigned, with each task allocated to exactly one picking robot.

(4) Different types of picking robots have different regional operation permissions.

(5) Picking robots depart from the depot sequentially at fixed time intervals and travel at a constant speed during task execution.

3.2. Problem modeling

Based on the problem defined above, the mathematical formulation is presented as follows:

Parameters:

S	Set of regions.
J	Set of task points, $J = \{J_0, J_1, J_2, \ldots, J_n\}$, where J_0 is the depot.
P	Number of robot types.
R	Set of picking robots, $R = \{R_{z,k} \mid z \in \{1, 2, \ldots, P\}, k \in K_z\}$, where P denotes the robot type scale and K_z represents the index set of type z robots.
J_r	Set of task points in region r, $J_r = \{J_j \mid S_j = r, j \in \{1, 2, \ldots, n\}\}$, where $r \in S$.
S_j	Region to which task point J_j belongs, $S_j \in S$.
$S_{z,k}$	Set of regions where picking robot $R_{z,k}$ can operate, $S_{z,k} \subseteq S$.
p_j	Weight to be picked at task point J_j.
P_j	Multi-region (MR) priority value of task point J_j, determined by picking efficiency, distance to the depot, and picking weight.
(x_j, y_j)	Coordinates of task point J_j.
n	Number of task points, excluding depot.
z	Picking robot type, $z \in \{1, 2, \ldots, P\}$.
k	Index of picking robot within its type, $k \in K_z = \{1, 2, \ldots, \lvert K_z \rvert\}$ for type z.
m	Total number of picking robots, $m = \sum_{z=1}^{P} \lvert K_z \rvert$.
t_0	Departure time of the first picking robot.
t_z	Average time for type z picking robot to pick one kilogram of crop.
$t_{z,k,0}$	Departure time of picking robot $R_{z,k}$.
$d_{jj'}$	Distance from task point J_j to task point $J_{j'}$, where $d(\cdot,\cdot)$ denotes the Manhattan distance between any two points. If $J_{j'}$ is the depot, $d_{jj'} = d\left(J_j, e^{\text{out}}_{S_j}\right) + d\left(e^{\text{out}}_{S_j}, J_0\right)$. If J_j is the depot, $d_{jj'} = d\left(J_0, e^{\text{in}}_{S_{j'}}\right) + d\left(e^{\text{in}}_{S_{j'}}, J_{j'}\right)$. If neither J_j nor $J_{j'}$ is the depot and they belong to the same region, $d_{jj'} = d\left(J_j, J_{j'}\right)$. If neither J_j nor $J_{j'}$ is the depot and they belong to different regions, $d_{jj'} = d\left(J_j, e^{\text{out}}_{S_j}\right) + d\left(e^{\text{in}}_{S_{j'}}, J_{j'}\right)$.
v	Travel speed of picking robots.
W	Maximum load of picking robots.
v_u	Unloading speed at depot.
L	Maximum number of rounds allowed for each picking robot, $L \leq n$.
$m_{z,k}$	Departure order index of robot $R_{z,k}$ among all robots, $m_{z,k} \in \{0, 1, \ldots, m-1\}$.
l	Round index of picking operation, $l \in \{1, 2, \ldots, L\}$.
Δt	Time interval between consecutive robot departures.
$PS_{r,r'}$	Fixed distance between region r and region r', where $r, r' \in S$.

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$(e_r^{\text{in}}, e_r^{\text{out}})$	Entrance and exit coordinates of region r, where $r \in S$.
ε	Workload deviation threshold for picking robots of the same type.
κ	Candidate list size for probabilistic robot selection.
Π	A complete solution representing all robot paths.

Decision Variables:

$x_{j,z,k}$	If picking robot $R_{z,k}$ executes task point J_j, $x_{j,z,k} = 1$; else $x_{j,z,k} = 0$.
$\gamma_{j,j',l,z,k}$	If $J_{j'}$ is the immediate successor of J_j in round l of picking robot $R_{z,k}$, $\gamma_{j,j',l,z,k} = 1$; else $\gamma_{j,j',l,z,k} = 0$.
$y_{j,l,z,k}$	If task point J_j is assigned to picking robot $R_{z,k}$ in round l, $y_{j,l,z,k} = 1$; else $y_{j,l,z,k} = 0$.
$e_{r,r',l,z,k}$	If picking robot $R_{z,k}$ in round l moves directly from region r to region r', $e_{r,r',l,z,k} = 1$; else $e_{r,r',l,z,k} = 0$.
$q_{j,l,z,k}$	If task point J_j is the last task of picking robot $R_{z,k}$ in round l, $q_{j,l,z,k} = 1$; else $q_{j,l,z,k} = 0$.
$\eta_{l,z,k}$	If any task point is assigned to picking robot $R_{z,k}$ in round l, $\eta_{l,z,k} = 1$; else $\eta_{l,z,k} = 0$.
$w_{j,l,z,k}$	Current load of picking robot $R_{z,k}$ after completing task J_j in round l, $0 \le w_{j,l,z,k} \le W$.
$C_{z,k}$	Completion time when picking robot $R_{z,k}$ returns to depot after executing all assigned tasks.

Objective:

$$\text{Minimize} \quad C_{max} = \max\{C_{z,k}\}$$

Subject to:

$$C_{max} \ge C_{z,k}, \quad \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z \tag{1}$$

$$\sum_{z=1}^{P} \sum_{k \in K_z} x_{j,z,k} = 1, \quad \forall j \in \{1,2,\ldots,n\} \tag{2}$$

$$x_{j,z,k} = 0, \quad \forall j \in \{1,2,\ldots,n\}, \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z \quad \text{if } S_j \notin S_{z,k} \tag{3}$$

$$\sum_{j=1}^{n} p_j \cdot y_{j,l,z,k} \le W, \quad \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \tag{4}$$

$$\begin{aligned} w_{j',l,z,k} \ge w_{j,l,z,k} + p_{j'} \cdot \gamma_{j,j',l,z,k} - W\left(1 - \gamma_{j,j',l,z,k}\right), \forall j \in \{0,1,\ldots,n\}, \\ \forall j' \in \{1,2,\ldots,n\}, j \ne j', \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \end{aligned} \tag{5}$$

$$w_{0,l,z,k} = 0, \quad \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \tag{6}$$

$$\sum_{j'=0}^{n} \gamma_{j,j',l,z,k} = y_{j,l,z,k}, \quad \forall j \in \{1,2,\ldots,n\}, \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \tag{7}$$

$$\begin{aligned} \sum_{j=0}^{n} \gamma_{j,j',l,z,k} = y_{j',l,z,k}, \quad \forall j' \in \{1,2,\ldots,n\}, \forall z \in \{1,2,\ldots,P\}, \\ \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \end{aligned} \tag{8}$$

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$$\sum_{j'=1}^{n} \gamma_{0,j',l,z,k} = \eta_{l,z,k}, \quad \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \tag{9}$$

$$\sum_{j=1}^{n} \gamma_{j,0,l,z,k} = \eta_{l,z,k}, \quad \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \tag{10}$$

$$e_{r,r',l,z,k} = \sum_{j \in J_r} \sum_{j' \in J_{r'}} \gamma_{j,j',l,z,k}, \quad \forall r, r' \in S, r \neq r', \forall l \in \{1,2,\ldots,L\}, \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z \tag{11}$$

$$\sum_{l=1}^{L} y_{j,l,z,k} = x_{j,z,k}, \quad \forall j \in \{1,2,\ldots,n\}, \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z \tag{12}$$

$$\sum_{j=1}^{n} q_{j,l,z,k} = \eta_{l,z,k}, \quad \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \tag{13}$$

$$\sum_{j=1}^{n} y_{j,l,z,k} \leq n \cdot \eta_{l,z,k}, \quad \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \tag{14}$$

$$\eta_{l,z,k} \leq \sum_{j=1}^{n} y_{j,l,z,k}, \quad \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \tag{15}$$

$$q_{j,l,z,k} \leq y_{j,l,z,k}, \quad \forall j \in \{1,2,\ldots,n\}, \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \tag{16}$$

$$q_{j,l,z,k} \leq \gamma_{j,0,l,z,k}, \quad \forall j \in \{1,2,\ldots,n\}, \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z, \forall l \in \{1,2,\ldots,L\} \tag{17}$$

$$\left| \sum_{j=1}^{n} p_j \cdot x_{j,z,k_1} - \sum_{j=1}^{n} p_j \cdot x_{j,z,k_2} \right| \leq \varepsilon \cdot \frac{\sum_{k \in K_z} \sum_{j=1}^{n} p_j \cdot x_{j,z,k}}{|K_z|}, \quad \forall z \in \{1,2,\ldots,P\}, \forall k_1, k_2 \in K_z, k_1 \neq k_2 \tag{18}$$

$$t_{z,k,0} = t_0 + \Delta t \cdot m_{z,k}, \quad \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z \tag{19}$$

$$C_{z,k} = t_{z,k,0} + \sum_{l=1}^{L} \left[\sum_{j=0}^{n} \sum_{j'=0, j \neq j'}^{n} \gamma_{j,j',l,z,k} \cdot \frac{d_{jj'}}{v} + \sum_{r \in S} \sum_{r' \in S, r \neq r'} e_{r,r',l,z,k} \cdot \frac{PS_{r,r'}}{v} + \sum_{j=1}^{n} y_{j,l,z,k} \cdot p_j \cdot t_z + \eta_{l,z,k} \cdot \frac{\sum_{j=1}^{n} p_j \cdot y_{j,l,z,k}}{v_u} \right], \quad \forall z \in \{1,2,\ldots,P\}, \forall k \in K_z \tag{20}$$

In the above mathematical model, Constraint (1) defines the Makespan as the maximum completion time of all picking robots. Constraint (2) ensures that each task point is assigned to exactly one picking robot. Constraint (3) ensures that picking robots can only execute task points within regions where they have operation permissions. Constraint (4) ensures that the total weight

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picked by each picking robot in each round does not exceed the maximum load. Constraints (5)-(6) specify the load update rule and initial load conditions of picking robots. Constraints (7)-(8) ensure that each assigned task point has exactly one predecessor and one successor, thereby guaranteeing path continuity. Constraints (9)-(10) ensure that, in each used round, the depot has exactly one predecessor and one successor. Based on the predecessor-successor relationships between task points, Constraint (11) determines whether a picking robot moves between regions in each round. Constraint (12) ensures that each task point assigned to a picking robot is executed in exactly one round of that robot. Constraint (13) specifies that if a round is used, exactly one unloading operation occurs at the depot in that round, and if the round is not used, no unloading operation occurs. Constraints (14)-(15) ensure that the value of $\eta_{l,z,k}$ remains consistent with the assignment outcome of task points in that round. Constraint (16) ensures that only a task point already assigned to picking robot $R_{z,k}$ in round l can be marked as the last task point of that round. Constraint (17) specifies that if a task point is marked as the last task point of that round, its successor must be the depot. Constraints (5)-(10) and (13)-(17) jointly establish that the picking robot departs from and returns to the depot in every round that is used. Regional operation permission constraints, maximum capacity constraints, and robot heterogeneity are not mutually independent but are instead coupled with one another, jointly determining both the feasibility of task assignment and the robot's completion time. Constraint (18) balances workloads among picking robots of the same type, thereby improving resource utilization efficiency. Constraint (19) coordinates departure times of picking robots to prevent initial collisions. Constraint (20) calculates the completion time of each picking robot, which consists of five main components: the initial departure time, travel time between task points, inter-region movement time, picking operation time, and unloading time at the depot after each round.

3.3. Example instance

To illustrate the MRHPRTAS problem being studied in this paper, this section presents a specific instance, consisting of 20 task points to be completed by 3 different types of picking robots. The system parameters are set as follows: $v = 72$, $v_u = 30$, $W = 150$, and $\Delta t = 0.0833$. The picking time coefficients for different types of robots are $t_1 = 0.50$, $t_2 = 0.53$, $t_3 = 0.58$.

According to constraint (19), the initial departure times for picking robots are $t_{1,1,0} = 0$, $t_{2,1,0} = 0.08$, and $t_{3,1,0} = 0.17$, respectively. The workload deviation threshold is set to $\varepsilon = 0.55$.

The task sequence for picking robot $R_{1,1}$ is: (1) $J_0 \to J_9 \to J_8 \to J_{18} \to J_5 \to J_{19} \to J_{16} \to J_0$; (2) $J_0 \to J_2 \to J_{20} \to J_7 \to J_0$.

The task sequence for picking robot $R_{2,1}$ is: (1) $J_0 \to J_{13} \to J_0$; (2) $J_0 \to J_{14} \to J_3 \to J_{11} \to J_0$; (3) $J_0 \to J_{12} \to J_1 \to J_0$.

The task sequence for picking robot $R_{3,1}$ is: (1) $J_0 \to J_6 \to J_4 \to J_0$; (2) $J_0 \to J_{15} \to J_{10} \to J_{17} \to J_0$.

Considering inter-region movements, taking the first round of picking robot $R_{1,1}$ as an example, its complete travel route is: $J_0 \to e_C^{\text{in}} \to J_9 \to J_8 \to J_{18} \to J_5 \to e_C^{\text{out}} \to e_A^{\text{in}} \to J_{19} \to J_{16} \to e_A^{\text{out}} \to J_0$.

Table 1. Time details for each picking robot's rounds.

Robot	Round	Departure time	Travel time	Picking time	Unloading time	Round completion time
$R_{1,1}$	1	0.00	4.31	69.50	4.63	78.44
	2	78.44	2.08	44.00	2.93	127.45
$R_{2,1}$	1	0.08	2.08	26.65	1.67	30.48
	2	30.48	1.67	31.98	2.00	66.13
	3	66.13	3.33	57.03	3.57	130.06
$R_{3,1}$	1	0.17	1.67	55.39	3.17	60.40
	2	60.40	4.44	59.47	3.40	127.71

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The travel distance for $R_{1,1}$ is $d_{0,e_C^{\text{in}}} + d_{e_C^{\text{in}},9} + d_{9,8} + d_{8,18} + d_{18,5} + d_{5,e_C^{\text{out}}} + PS_{C,A} + d_{e_A^{\text{in}},19} + d_{19,16} + d_{16,e_A^{\text{out}}} + d_{e_A^{\text{out}},0} = 310$, and the total travel time is $310/72 = 4.31$. The picking time is $(p_9 + p_8 + p_{18} + p_5 + p_{19} + p_{16}) \cdot t_1 = 139 \times 0.50 = 69.50$. The unloading time is $(p_9 + p_8 + p_{18} + p_5 + p_{19} + p_{16})/v_u = 139/30 = 4.63$. Therefore, the completion time of the first round for $R_{1,1}$ is $t_{1,1,0} + 4.31 + 69.50 + 4.63 = 78.44$. Through similar calculations, the detailed information for each picking robot can be obtained, as shown in Table 1. According to constraint (1), the maximum completion time of the system is:

$$C_{max} = \max(C_{1,1}, C_{2,1}, C_{3,1}) = \max(127.45, 130.06, 127.71) = 130.06$$

4. Proposed algorithm

This paper proposes the MRHDABC algorithm, derived from the DABC framework, to address the MRHPRTAS problem in smart farms. The MRHDABC algorithm first employs four methods: the MRNEH-based heuristic, the MRNEHIG-based heuristic, the IG-based heuristic, and random initialization, to generate high-quality initial solutions, and then enters an iterative process comprising the employed bee phase, the onlooker bee phase, and the scout bee phase, as illustrated in Fig. 2. The pseudocode of the proposed MRHDABC algorithm is given in Algorithm 1.

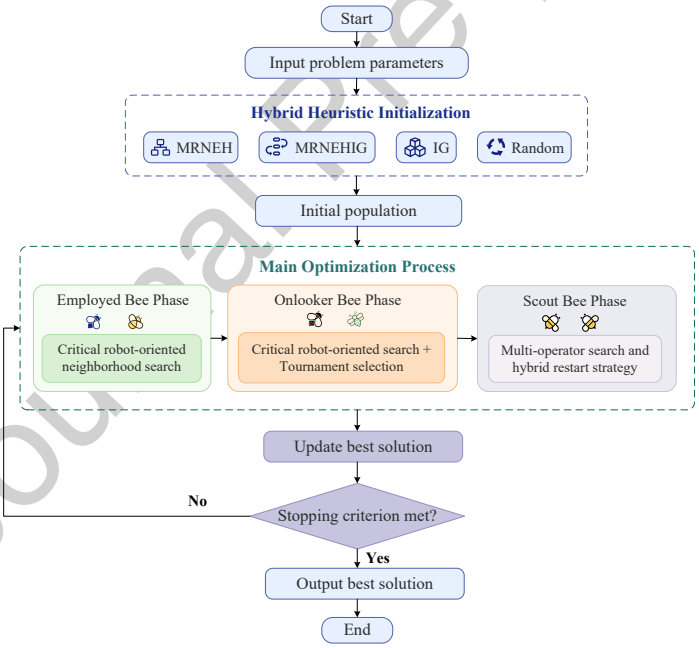


Fig. 2. Framework of the proposed MRHDABC algorithm.

Algorithm 1. MRHDABC

Input: Population size $Psize$, local search depth β, exploration limit threshold λ
Output: Global best solution Π_{best}

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1: Generating $Psize$ initial individuals using MRNEH-based, MRNEHIG-based, IG-based heuristics and random initialization
2: **while** the stopping criterion is not met **do**
3: The employed bee phase // Section 4.4
4: The onlooker bee phase // Section 4.5
5: The scout bee phase // Section 4.6
6: Update Π_{best}
7: **end while**

4.1. Solution representation

This paper designs a hierarchical solution encoding method based on round sequences to represent the task allocation and execution schemes of heterogeneous picking robots. In the task assignment process, whenever inserting the next task into the current round would cause the cumulative picked weight to exceed the maximum load, the algorithm opens a new round for the robot to accommodate the remaining tasks, and the number of rounds actually executed by the robot is thus dynamically determined by Constraint (4). Let the smart farm contain n task points and P types of heterogeneous picking robots, where K_z robots are configured for type z. The complete solution is represented as $\Pi = [\pi_{1,1}, \pi_{1,2}, \ldots, \pi_{1,K_1}, \ldots, \pi_{P,1}, \pi_{P,2}, \ldots, \pi_{P,K_P}]$, where $\pi_{z,k}$ represents the task execution scheme of $R_{z,k}$, defined as $\pi_{z,k} = [\tau_1, \tau_2, \ldots, \tau_L]$. $\tau_l = [\tau_{l,1}, \tau_{l,2}, \ldots, \tau_{l,h}]$ represents the task sequence of $R_{z,k}$ in the l-th round, where h is the number of tasks in that round. Based on the specific example in Section 3.3, the complete solution is represented as $\Pi = [\pi_{1,1}, \pi_{2,1}, \pi_{3,1}]$, where $\pi_{1,1} = [[9, 8, 18, 5, 19, 16], [2, 20, 7]]$ indicates that $R_{1,1}$ executes 2 rounds, $\pi_{2,1} = [[13], [14, 3, 11], [12, 1]]$ indicates that $R_{2,1}$ executes 3 rounds, and $\pi_{3,1} = [[6, 4], [15, 10, 17]]$ indicates that $R_{3,1}$ executes 2 rounds. Taking $R_{3,1}$ as an example, the task point visiting sequence in the first round is $6 \rightarrow 4$, and the task point visiting sequence in the second round is $15 \rightarrow 10 \rightarrow 17$. Fig. 3 shows the path distribution of $R_{3,1}$.

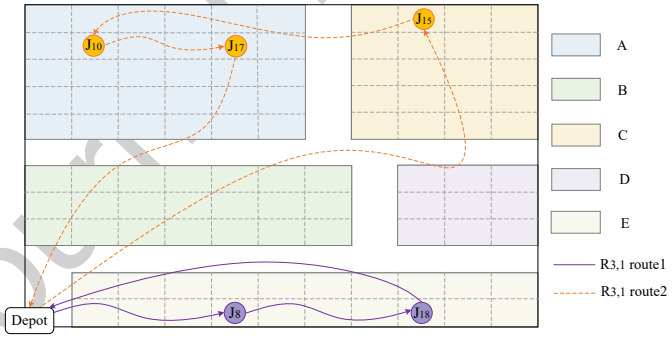


Fig. 3. Distribution of $R_{3,1}$ routes.

4.2. Proposed heuristics and initialization

Given the NP-hard nature of the MRHPRTAS problem and its multi-region heterogeneous constraint coupling characteristics, traditional random initialization strategies suffer from limitations of low feasible solution generation rate and poor solution quality. This paper constructs a multi-strategy hybrid heuristic initialization framework for multi-region heterogeneous robots, which efficiently generates high-quality feasible solutions through the integration of three heuristic algorithms: MRNEH, MRNEHIG, and IG.

The MRNEH-based heuristic algorithm generates one high-quality initial solution, employs MR multi-factor priority calculation for task sorting, and sequentially inserts tasks into optimal

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positions to minimize C_{max}. The algorithm randomly selects one from the top κ ($\kappa = 3$) feasible picking robots with minimum workload for task assignment, with its pseudocode shown in Algorithm 2. The MRNEHIG-based heuristic algorithm generates approximately 1/4 of the initial population and adopts a region grouping strategy. It first randomly determines the region processing order, then sorts tasks within each region according to MR priority, and achieves optimal allocation of intra-region tasks through optimal position insertion, with its pseudocode shown in Algorithm 3. The IG-based heuristic algorithm generates approximately 1/4 of the initial population and adopts a region-oriented random construction strategy. It first randomly determines the region processing order and the task sequence within each region, then finds the feasible position that minimizes C_{max} through greedy position search, with its pseudocode shown in Algorithm 4. The remaining initial solutions are generated through random initialization methods to ensure population diversity.

Algorithm 2. MRNEH-based heuristic algorithm

Input: Task set J, picking robot set R
Output: Complete solution Π

begin
  for $j = 1$ to n **do**
    Calculate multi-dimensional MR priority value P_j for task J_j
    Apply random perturbation to P_j
  end for
  $TaskSeq \leftarrow$ Sort tasks in descending order by P_j values
  $TaskSeq \leftarrow$ Perform adaptive task swapping operation
  $\Pi \leftarrow \emptyset$
  for $i = 1$ to n **do**
    $J_i \leftarrow TaskSeq[i]$
    $R_{sorted} \leftarrow$ Obtain feasible robots for J_i and sort by current workload
    $R_{selected} \leftarrow$ Randomly select from top κ robots with minimum workload
    if feasible(J_i, $R_{selected}$) **then**
      insert J_i at optimal position in Π
    else
      select next candidate robot
    end if
  end for
  if infeasible(Π) **then**
    $\Pi \leftarrow$ Execute constraint repair operation
  end if
  return Π
end

Algorithm 3. MRNEHIG-based heuristic algorithm

Input: Task set J, picking robot set R, region set S, maximum load W
Output: Complete solution Π

begin
  for $r \in S$ **do**
    $J_r \leftarrow$ Extract tasks in region r
  end for
  $S' \leftarrow$ Randomize region processing sequence
  $\Pi \leftarrow \emptyset$

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```
    for r ∈ S′ do
        for j ∈ J_r do
            Calculate multi-dimensional MR priority value P_j for task j
            Apply random perturbation to P_j
        end for
        Group_r ← Sort J_r in descending order by P_j
        R_r ← Obtain executable robot set for region r
        for j ∈ Group_r do
            Sort R_r by current workload
            R_selected ← Probabilistic selection from R_r
            Insert task j at optimal round position in R_selected
            Update Π
        end for
    end for
    return Π
end
```

Algorithm 4. IG-based heuristic algorithm

Input: Task set J, picking robot set R, region set S, maximum load W
Output: Complete solution Π

```
begin
    S′ ← Randomize region processing sequence
    Π ← ∅
    for r ∈ S′ do
        J_r′ ← Randomize task sequence within region r
        R_r ← Obtain executable robot set for region r
        for j ∈ J_r′ do
            R_r′ ← Randomize robot order in R_r
            assigned ← Round-aware greedy insertion(j, R_r′, Π)
            if assigned = false then
                return ∅
        end for
    end for
    return Π
end
```

4.3. Local search strategy

Insertion, exchange, and crossover operators are widely applied neighborhood search methods in scheduling problems. However, when facing the complex constraint conditions of the MRHPRTAS problem, traditional neighborhood operators fail to simultaneously satisfy regional operating permission constraints and maximum capacity constraint, lacking effective multi-constraint coupling handling capability. Therefore, this paper designs six targeted neighborhood operators, whose implementation mechanisms are shown in Fig. 4:

(1) Single-robot insertion operation (SR-Insert)

Randomly select a picking robot $R_{z,k}$ and its task execution scheme $\pi_{z,k}$ from the current solution Π. Randomly select a round task sequence $\tau_l = [\tau_{l,1}, \tau_{l,2}, \ldots, \tau_{l,h}]$, and randomly select a task $\tau_{l,i}$ and an insertion position i' from the sequence ($\forall i, i' \in \{1, 2, \ldots, h\}$ and $i \neq i'$). Remove task $\tau_{l,i}$ from its original position and insert it into position i', forming a new task sequence τ'_l, as shown in Fig. 4(a).

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(2) Single-robot exchange operation (SR-Exchange)

Randomly select a picking robot $R_{z,k}$ and its task execution scheme $\pi_{z,k}$ from the current solution Π. Randomly select two different tasks $\tau_{l,i}$ and $\tau_{l,j}$ from a round task sequence $\tau_l = [\tau_{l,1}, \tau_{l,2}, \ldots, \tau_{l,h}]$ ($\forall i, j \in \{1, 2, \ldots, h\}$ and $i \neq j$), and exchange the execution order of task $\tau_{l,i}$ and task $\tau_{l,j}$, forming a new task sequence τ'_l, as shown in Fig. 4(b).

(3) Single-robot crossover operation (SR-Cross)

Randomly select a picking robot $R_{z,k}$ and its task execution scheme $\pi_{z,k}$ from the current solution Π. Randomly select a round task sequence $\tau_l = [\tau_{l,1}, \tau_{l,2}, \ldots, \tau_{l,h}]$, and randomly select two crossover points i and j in the sequence ($\forall i, j \in \{1, 2, \ldots, h\}, i \leq j$). Reverse the task sequence between positions i and j, forming a new task sequence τ'_l, as shown in Fig. 4(c).

(4) Multi-robot insertion operation (MR-Insert)

Randomly select two different picking robots $R_{z,k}$ and $R_{z',k'}$ and their task execution schemes $\pi_{z,k}$ and $\pi_{z',k'}$ from the current solution Π. Randomly select a task $\tau_{l,i}$ from a round task sequence $\tau_l = [\tau_{l,1}, \tau_{l,2}, \ldots, \tau_{l,h}]$ of $R_{z,k}$ ($\forall i \in \{1, 2, \ldots, h\}$), and randomly select an insertion position i' from a round task sequence $\tau_{l'} = [\tau_{l',1}, \tau_{l',2}, \ldots, \tau_{l',h'}]$ of $R_{z',k'}$ ($\forall i' \in \{1, 2, \ldots, h'\}$). Remove task $\tau_{l,i}$ from sequence τ_l and insert it into sequence $\tau_{l'}$ at position i', as shown in Fig. 4(d).

(5) Multi-robot exchange operation (MR-Exchange)

Randomly select two different picking robots $R_{z,k}$ and $R_{z',k'}$ from the current solution Π. Randomly select one task from each of their task sequences: select task $\tau_{l,i}$ from $\tau_l = [\tau_{l,1}, \tau_{l,2}, \ldots, \tau_{l,h}]$ ($\forall i \in \{1, 2, \ldots, h\}$), and select task $\tau_{l',j}$ from $\tau_{l'} = [\tau_{l',1}, \tau_{l',2}, \ldots, \tau_{l',h'}]$ ($\forall j \in \{1, 2, \ldots, h'\}$). Exchange task $\tau_{l,i}$ and task $\tau_{l',j}$, as shown in Fig. 4(e).

(6) Multi-robot crossover operation (MR-Cross)

Randomly select two different picking robots $R_{z,k}$ and $R_{z',k'}$ from the current solution Π, and determine their task sequences $\tau_l = [\tau_{l,1}, \tau_{l,2}, \ldots, \tau_{l,h}]$ and $\tau_{l'} = [\tau_{l',1}, \tau_{l',2}, \ldots, \tau_{l',h'}]$. The operation randomly selects crossover points i and j in the two sequences respectively, and exchanges the task segments after the crossover points in both sequences, forming new task sequences τ'_l and $\tau'_{l'}$. During the exchange process, the operation ensures that each task appears only once, as illustrated in Fig. 4(f).

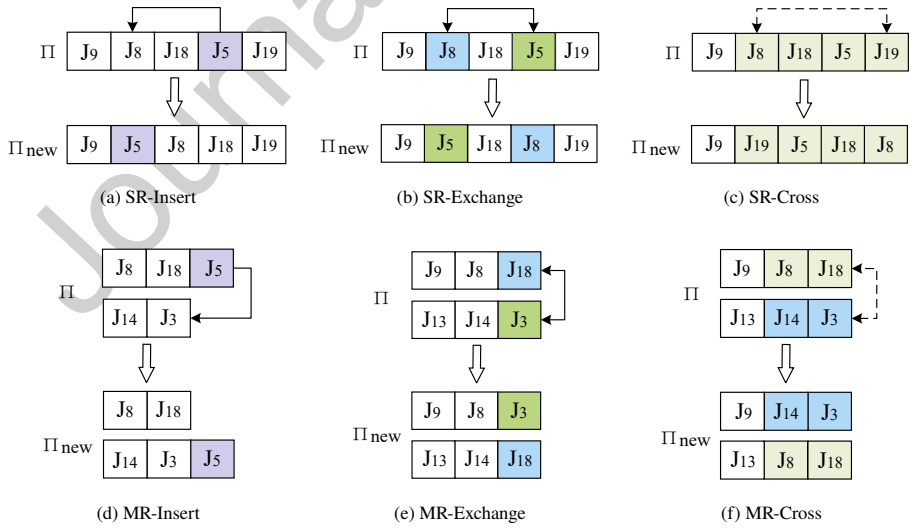


Fig. 4. Neighborhood operators.

Each neighborhood operator performs a constraint check before the new solution is accepted.

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For SR-Insert, SR-Exchange, and SR-Cross, no task is transferred across picking robots. After generating a new task sequence, these operators only verify whether it satisfies the maximum capacity constraint. If not, the operation is discarded and the solution remains unchanged. For MR-Insert, MR-Exchange, and MR-Cross, a task is transferred across picking robots. These operators verify whether the new solution simultaneously satisfies the regional operating permission constraint and the maximum capacity constraint. If not, the operation is discarded and the solution remains unchanged. After the constraint check is passed, for these single-robot operators, the algorithm only recalculates the completion time of the selected robot. For these multi-robot operators, the algorithm only recalculates the completion times of the two selected robots. The task execution schemes of the remaining robots remain unchanged.

4.4. The employed bee phase

The employed bee phase is responsible for local search and solution improvement on current food sources. Each food source Π_i is associated with a non-improvement counter $trials(\Pi_i)$, which records the number of consecutive iterations without improvement. The algorithm identifies the robot with the longest completion time as the critical robot $R_{critical}$, whose completion time determines the system C_{max}. The algorithm randomly selects one of the six neighborhood operators defined in Section 4.3 and performs at most β neighborhood search operations on each food source. Each of the six neighborhood operators is applied to randomly selected picking robots and is not restricted to the critical robot. Once $C_{max}(\Pi_{new}) < C_{max}(\Pi_i)$ is found, the algorithm immediately replaces the current solution with the new solution and resets $trials(\Pi_i)$ to 0, while terminating the subsequent search of the current food source. To improve computational efficiency, the algorithm only recalculates the completion times of the critical robot and the picking robots affected by the neighborhood operation. If a food source fails to obtain improvement after β search operations, its $trials(\Pi_i)$ counter is incremented by 1. Algorithm 5 presents the complete pseudocode for the employed bee phase.

Algorithm 5. The employed bee phase

Input: Population Π_i, local search depth β
Output: Updated population Π_i

```
1:  begin
2:      for i = 1 to Psize do
3:          Π_new ← Π_i ; flag = false
4:          for j = 1 to β do
5:              R_critical ← identify the critical robot
6:              O = rand() % 6
7:              switch (O)
8:                  case 0: Execute the SR-Insert on Π_new
9:                  case 1: Execute the SR-Exchange on Π_new
10:                 case 2: Execute the SR-Cross on Π_new
11:                 case 3: Execute the MR-Insert on Π_new
12:                 case 4: Execute the MR-Exchange on Π_new
13:                 case 5: Execute the MR-Cross on Π_new
14:             end switch
15:             R_affected ← Set of robots affected by the operation
16:             if C_max(Π_new) < C_max(Π_i) then
17:                 Π_i ← Π_new ; trials(Π_i) ← 0 ; flag = true
18:                 break
19:             end if
20:         end for
```

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```
        if flag = false then
            trials(Π_i)++
        end if
    end for
end
```

4.5. The onlooker bee phase

Traditional DABC algorithms employ fitness proportional selection mechanisms, which easily fall into local optima and lack problem-specific search strategies. The MRHDABC algorithm introduces the concept of critical robots, defining the robot with the longest completion time as the critical robot, whose completion time equals the system Makespan. The algorithm designs six critical robot-oriented neighborhood search operators, ensuring at least one operation target is a critical robot to specifically optimize system bottlenecks. The algorithm dynamically identifies critical robots, enabling the optimization process to respond to system bottleneck changes in real-time. Unlike traditional full solution evaluation, MRHDABC adopts a local evaluation strategy, only recalculating the completion times of affected robots. The algorithm employs a tournament selection mechanism to replace fitness proportional selection, effectively avoiding local optima. The tournament selection mechanism randomly selects two solutions from the population, and the one with the smaller C_{max} is chosen as $\Pi_{selected}$. When the new solution's $C_{max}(\Pi_{new})$ is superior to the worst solution's $C_{max}(\Pi_{worst})$ in the population, replacement is performed and $trials(\Pi_{worst})$ is reset to 0; otherwise, $trials(\Pi_{worst})$ is incremented. Because the tournament selection mechanism does not rely on the fitness distribution of the entire population, the algorithm is able to continuously explore different regions of the solution space across subsequent iterations, thereby enhancing its global search capability. Algorithm 6 provides the complete pseudocode for the onlooker bee phase.

Algorithm 6. The onlooker bee phase

Input: Population Π_i
Output: Updated population Π_i

```
begin
    for i = 1 to Psize do
        Π_worst ← select the worst individual from the population
        Π_selected ← select the better one from two randomly chosen individuals
        R_critical ← identify the critical robot
        O = rand() % 6 ; Π_new ← Π_selected
        switch (O)
            case 0: Execute the KSR-Insert on Π_new
            case 1: Execute the KSR-Exchange on Π_new
            case 2: Execute the KSR-Cross on Π_new
            case 3: Execute the KMR-Insert on Π_new
            case 4: Execute the KMR-Exchange on Π_new
            case 5: Execute the KMR-Cross on Π_new
        end switch
        if C_max(Π_new) < C_max(Π_worst) then
            Π_worst ← Π_new ; trials(Π_worst) ← 0
        else
            trials(Π_worst)++
        end if
    end for
end
```

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4.6. The scout bee phase

The scout bee phase is responsible for handling unimproved solutions and maintaining population diversity. The random replacement strategy in traditional DABC algorithms tends to lose useful information. When a food source fails to achieve improvement over consecutive λ iterations, the MRHDABC algorithm employs multi-operator collaborative search for solution reconstruction. The algorithm randomly selects several critical robot-oriented operators, as well as several neighborhood operators defined in Section 4.3, in order to specifically optimize system bottlenecks. It then tries these operators in sequence. Once an attempt yields $C_{max}(\Pi_{new})$ superior to $C_{max}(\Pi_i)$, the corresponding solution is adopted, and no further operators are tried. To improve computational efficiency, the algorithm adopts a critical robot-oriented incremental evaluation strategy, recalculating only the completion times of robots affected by neighborhood operations. When all operator attempts fail to achieve improvement, the algorithm executes a hybrid restart strategy, combining random restart and guided restart. In the guided restart, the algorithm uses the global best solution as the foundation and avoids premature convergence to local optima through random perturbation of partial task assignments. The random restart generates an entirely new task assignment for the unimproved food source, while the guided restart retains the task assignments in the global best solution that are not perturbed. Together, they help maintain population diversity while retaining useful information from the global best solution, allowing the algorithm to escape prolonged non-improvement. Algorithm 7 presents the pseudocode for the scout bee phase.

In summary, the employed bee phase improves each food source through general local search and an acceptance criterion based on C_{max}. The onlooker bee phase specifically optimizes system bottlenecks through the critical robot-oriented operators, and enhances the algorithm's global search capability through the tournament selection mechanism. The scout bee phase combines random restart and guided restart to continuously maintain population diversity. In conclusion, the global search capability and diversity of the MRHDABC algorithm are notably enhanced.

Algorithm 7. The scout bee phase

Input: Population Π_i, exploration limit threshold λ, global best solution Π_{best}
Output: Updated population Π_i

begin
  for i = 1 to $Psize$ **do**
    if $trials(\Pi_i) > \lambda$ **then**
      $\Pi_{new} \leftarrow \Pi_i$; $flag$ = false
      Execute multi-operator integrated search on Π_{new}
      if $C_{max}(\Pi_{new}) < C_{max}(\Pi_i)$ **then**
        $\Pi_i \leftarrow \Pi_{new}$; $trials(\Pi_i) \leftarrow 0$; $flag$ = true
      end if
      if $flag$ = false **then**
        $\Pi_i \leftarrow$ hybrid restart strategy(Π_{best})
        $trials(\Pi_i) \leftarrow 0$
      end if
    end if
  end for
end

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4.7. Computational Complexity of MRHDABC Algorithm

This section analyzes the time complexity of the MRHDABC algorithm. Let PS denote the population size, n denote the number of picking task points, m denote the number of heterogeneous picking robots, S denote the number of operational regions, $MaxG$ denote the maximum number of iterations, and β denote the local search depth parameter.

The complexity of objective function evaluation for a single solution is $O(nm + mS^2)$. In the initialization phase, the complexities of MRNEH heuristic, MRNEHIG heuristic, IG heuristic, and random heuristic are $O(n^2m)$, $O(Sn^2m)$, $O(n^2m)$, and $O(nm)$ respectively, thus the total initialization time complexity is $O(PS \cdot Sn^2m)$.

In the main iterative process of the algorithm, the time complexity of neighborhood search operators is $O(nm)$. Based on the complexity of neighborhood search operators, the employed bee phase performs at most β neighborhood searches for each food source, with a time complexity of $O(PS \cdot nm \cdot \beta)$. The onlooker bee phase and scout bee phase both have a time complexity of $O(PS \cdot nm)$. The population update operation requires sorting PS individuals by fitness values, with a time complexity of $O(PS \cdot \log(PS))$. The time complexity of a single iteration is $O(PS \cdot nm \cdot \beta + PS \cdot \log(PS))$.

Given that the algorithm can execute at most $MaxG$ iterations within the specified time limit, the total time complexity of the iterative part is $O(MaxG \cdot (PS \cdot nm \cdot \beta + PS \cdot \log(PS)))$. Combining the initialization and iterative processes, the total time complexity of the MRHDABC algorithm is $O(PS \cdot Sn^2m + MaxG \cdot (PS \cdot nm \cdot \beta + PS \cdot \log(PS)))$.

This analysis indicates that the time complexity of the MRHDABC algorithm is mainly influenced by the population size PS, the number of task points n, the number of robots m, the number of operational regions S, and the local search depth parameter β.

5. Numerical results and comparison

In this section, we first calibrate the parameters of the proposed MRHDABC algorithm using Design of Experiments (DOE) and Analysis of Variance (ANOVA)[35, 36].

5.1. Test instances

To evaluate the performance of the MRHDABC algorithm, we use 100 test instances, divided into four categories (20, 40, 60, 80 task points). Tasks are distributed across 5 regions, with picking weights randomly generated within the range of [5, 60]. The picking robot configuration $\{K_1, K_2, K_3\}$ varies according to task scale: $\{K_1 = 1, K_2 = 1, K_3 = 1\}$ configuration for 20 task points, $\{K_1 = 2, K_2 = 2, K_3 = 1\}$ configuration for 40 task points, $\{K_1 = 3, K_2 = 1, K_3 = 2\}$ configuration for 60 task points, and $\{K_1 = 3, K_2 = 3, K_3 = 3\}$ configuration for 80 task points. System parameters: warehouse location at $(0, 0)$, $W = 150\,\text{kg}$, $v = 72\,\text{m/min}$, $v_u = 30\,\text{kg/min}$, $\Delta t = 0.0833\,\text{min}$. Detailed information about different types of picking robots is presented in Table 2. For each task scale, five test instances are randomly selected, yielding a total of 20 test instances that constitute the calibration set, while the remaining 80 test instances constitute the test set.

Table 2. Picking robot type information.

Type z	Picking time coefficient t_z	Permitted operating regions $S_{z,k}$
1	0.500	A, B, C, D
2	0.533	B, D, E
3	0.583	A, C, E

Note: The unit of picking time coefficient is min/kg.

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5.2. Experimental design

All algorithms are implemented in C++ and executed on an Intel i7-10700 CPU @ 2.90GHz with 8.00GB RAM running Windows 11 operating system. The termination condition for all experiments is the maximum CPU consumption time, set to $0.1 \cdot n \cdot m$ seconds. The value of this coefficient is determined with reference to similar termination criteria adopted in related studies [5, 7, 23], together with the actual solution-time requirements in smart farms. Each test instance is run independently 10 times to ensure result stability. The Relative Percentage Increase (RPI) is used as the response variable, calculated as shown in Equation (21).

$$RPI = \frac{C_{alg} - C_{best}}{C_{best}} \times 100\% \tag{21}$$

where C_{alg} is the Makespan obtained by a given algorithm, and C_{best} is the minimum Makespan found among all compared algorithms.

To reduce the influence of algorithmic randomness on the results, the mean value of RPI over N independent runs is defined as the Average Relative Percentage Increase (ARPI), calculated as shown in Equation (22).

$$ARPI = \frac{1}{N} \sum_{i=1}^{N} RPI_i \tag{22}$$

where RPI_i is the RPI value obtained in the i-th independent run, and N is the number of independent runs performed on each test instance.

To further evaluate solution stability, the standard deviation of RPI over N independent runs is calculated as shown in Equation (23).

$$Std(RPI) = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (RPI_i - ARPI)^2} \tag{23}$$

5.3. Parameter tuning

All parameter calibration experiments in this section are conducted using the calibration set. For each parameter value, each test instance is run independently 10 times.

5.3.1. Model parameter calibration

The workload deviation threshold ε in Constraint (18) bounds the relative deviation between the workloads of picking robots of the same type. This section analyzes the effect of ε on algorithm performance, with $\varepsilon \in \{0.40, 0.45, 0.50, 0.55, 0.60\}$. Solution quality is evaluated using ARPI. Workload balance is evaluated using the coefficient of variation (CV), calculated as shown in Equation (24). A smaller CV indicates a more balanced workload.

$$CV = \frac{1}{P} \sum_{z=1}^{P} \frac{Std\left(\left\{\sum_{j=1}^{n} p_j x_{j,z,k}\right\}_{k \in K_z}\right)}{Mean\left(\left\{\sum_{j=1}^{n} p_j x_{j,z,k}\right\}_{k \in K_z}\right)} \times 100\% \tag{24}$$

where $\sum_{j=1}^{n} p_j x_{j,z,k}$ denotes the total workload of picking robot $R_{z,k}$, and $Std(\cdot)$ and $Mean(\cdot)$ denote the standard deviation and mean operators, respectively.

As shown in Table 3 and Fig. 5, both ARPI and CV reach their minimum values at $\varepsilon = 0.55$. Considering both indicators, $\varepsilon = 0.55$ is adopted as the workload deviation threshold in this paper.

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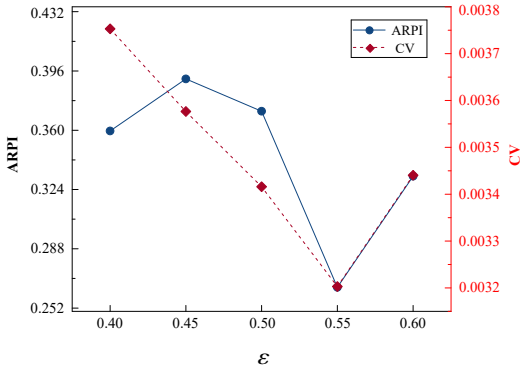


Fig. 5. ARPI and average CV of MRHDABC under different values of ε.

Table 3. Sensitivity analysis results of the workload deviation threshold ε.

ε	ARPI	Average CV
0.40	0.36%	0.38%
0.45	0.39%	0.36%
0.50	0.37%	0.34%
0.55	0.26%	0.32%
0.60	0.33%	0.34%

5.3.2. Algorithm parameter calibration

To determine the optimal parameter combination for the MRHDABC algorithm, we employ orthogonal experimental design for systematic parameter tuning. Orthogonal experimental design is an efficient method that can explore the influence of multi-factor multi-level parameters with minimal experimental trials. Based on the characteristics of the MRHDABC algorithm, three key parameters are selected for tuning: colony size ($Psize$), exploration limit (λ), and local search depth (β). Each parameter is set with four levels, as shown in Table 4. We adopt the orthogonal table ($L_{16}(4^3)$) to design experiments. Solution quality is evaluated using ARPI, as defined in Section 5.2. Table 5 shows the response values and significance ranking of each parameter at different levels.

According to the range analysis, the order of influence of each parameter on algorithm performance is: $Psize > \lambda > \beta$. Fig. 6 shows the orthogonal experimental analysis results, where $Psize$, λ, and β all achieve their best performance at the second level. Therefore, the final parameter settings for the MRHDABC algorithm are: $Psize = 80$, $\lambda = 150$, and $\beta = 5$.

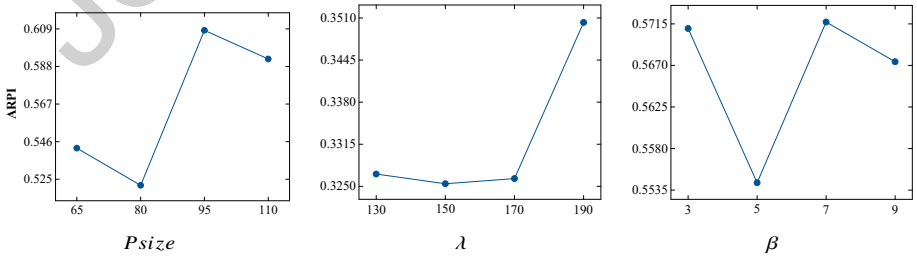


Fig. 6. Factor level trend.

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Table 4. Combinations of parameter values.

Parameter	Level			
	1	2	3	4
$Psize$	65	80	95	110
λ	130	150	170	190
β	3	5	7	9

Table 5. Response value and significance rank.

Level	Factor		
	$Psize$	λ	β
1	0.5424	0.3269	0.5710
2	0.5216	0.3254	0.5543
3	0.6082	0.3262	0.5717
4	0.5922	0.3503	0.5674
Delta	0.0866	0.0249	0.0174
Rank	1	2	3

5.4. Component effectiveness analysis

To evaluate the effectiveness of each key component in the MRHDABC algorithm, this study conducts comparative analysis with three variant algorithms. These variant algorithms are: (1) MRHDABC_1: removing the heuristic initialization strategy and employing random initialization; (2) MRHDABC_2: removing the round-based load balancing strategy during initialization and employing random robot selection; (3) MRHDABC_3: removing the local search operators. All algorithms adopt the optimal parameter settings determined in Section 5.3 and employ the same termination criteria.

Fig. 7 shows the ARPI values and 95% confidence intervals for each algorithm variant. As can be seen from Fig. 7, MRHDABC obtains the smallest ARPI value, exhibiting the best performance. The confidence intervals of MRHDABC and MRHDABC_1 have no overlap, indicating that the proposed heuristic initialization contributes to generating high-quality initial solutions. The confidence intervals of MRHDABC and MRHDABC_2 have no overlap, demonstrating that the round-based load balancing strategy effectively promotes workload balance among robots of the same type and improves solution quality. The confidence intervals of MRHDABC and MRHDABC_3 also have no overlap, validating that the local search operators further improve solution quality. In summary, the experimental results verify the effectiveness of each key component in the MRHDABC algorithm.

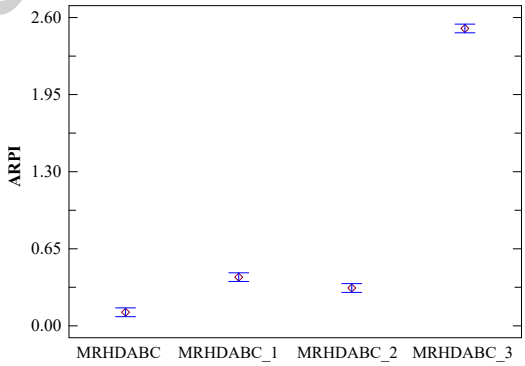


Fig. 7. 95% LSD confidence intervals for various MRHDABC variants.

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5.5. Comparison of metaheuristics

To evaluate MRHDABC's performance, it is compared with four state-of-the-art metaheuristic algorithms: Multi-objective Discrete Artificial Bee Colony (MODABC) [7], Rank-based Ant Colony Optimization (ACO-RNK) [14], Hybrid Tabu Search Multi-objective Genetic Algorithm (HTSMOGA) [20], and Adaptive Multi-objective Evolutionary Algorithm (AMOEA) [11]. To ensure fair comparison, all algorithms are configured with problem-specific parameters (Table 6). Each algorithm is executed independently 10 times on the 80 test instances, with the termination criterion set to $0.1 \cdot n \cdot m$ seconds.

Table 7 presents ARPI performance across varying problem scales. MRHDABC achieves an ARPI of 0.0150, significantly outperforming MODABC(2.4910), ACO-RNK(4.2333), HTSMOGA(7.0931), and AMOEA(3.4610). The performance gap between MRHDABC and the compared algorithms widens progressively with increasing problem scale, demonstrating MRHDABC's superior scalability.

ANOVA confirms statistically significant differences among algorithms (Fig. 8). Non-overlapping 95% LSD confidence intervals between MRHDABC and all compared algorithms confirm the statistical significance of MRHDABC's superior performance. Interaction analysis (Fig. 9) demonstrates MRHDABC's consistent superiority across all problem configurations. This superiority is attributed to critical robot-oriented search and load balancing mechanisms, which collectively exploit the problem's inherent structural properties.

Table 8 presents the standard deviation of RPI for each algorithm, used to evaluate solution stability. MRHDABC achieves the smallest standard deviation across all four problem scales, whereas the standard deviations of the other algorithms generally increase with problem scale, indicating that MRHDABC exhibits stronger solution stability.

Table 6. Calibration results of algorithm parameters.

Algorithms	Parameter settings
MRHDABC	$Psize = 80$, $\lambda = 150$, $\beta = 5$
ACO-RNK	$\bar{A} = 50$, $\delta = 1.0$, $\theta = 2.0$, $\psi = 2.0$, $\rho = 0.1$, $a = 0.1$
AMOEA	$ps = 40$, $pc = 0.9$, $\theta = 10$
HTSMOGA	$N = 100$, $T = 100$, $Pc = 0.5$, $pm = 0.5$
MODABC	$Psize = 30$, $TN = 3$, $g = 50$

Table 7. RPI values of algorithms under different tasks.

n-m	MRHDABC	MODABC	ACO-RNK	HTSMOGA	AMOEA
20-3	**0.0599**	0.2810	0.4295	1.1824	1.2838
40-5	**0.0000**	1.7535	3.2958	5.9538	3.2171
60-6	**0.0000**	2.2295	3.9166	6.9419	3.7308
80-9	**0.0000**	5.7000	9.2914	14.2943	5.6122
ARPI	**0.0150**	2.4910	4.2333	7.0931	3.4610

Table 8. Stability analysis of algorithms under different tasks.

n-m	MRHDABC	MODABC	ACO-RNK	HTSMOGA	AMOEA
20-3	0.0985	0.2408	0.2708	0.5226	0.5947
40-5	0.0000	0.4991	0.8447	1.5445	0.9431
60-6	0.0000	0.5324	0.7815	1.5147	0.9689
80-9	0.0000	1.1783	1.4637	2.1947	1.2720
Overall	0.0246	0.6126	0.8402	1.4441	0.9446

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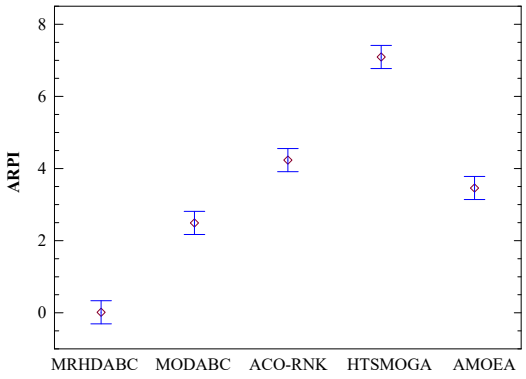


Fig. 8. Mean plots with 95% LSD confidence intervals for all comparison algorithms.

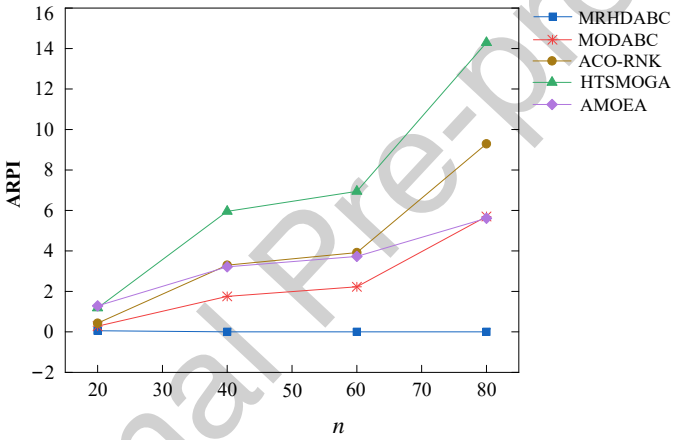


Fig. 9. The means plots of interaction with 95% Tukey HSD intervals.

To examine the statistical significance of the performance differences among algorithms, the Wilcoxon signed-rank test is employed to compare the paired RPI results of MRHDABC and each compared algorithm across the 80 test instances, with the results presented in Table 9. Across all problem scales, the p-values for MRHDABC versus each compared algorithm are all below 0.001, indicating that MRHDABC's performance advantage is statistically significant.

Fig. 10 presents the convergence curves of the five algorithms for the $n = 40$ and $n = 60$ scales, with the vertical axis representing the average C_{max}. MRHDABC converges faster than the other algorithms, and its objective function value ultimately stabilizes at a lower level, indicating that MRHDABC reaches its best achievable solution within the given computation time.

The Friedman test [37] further confirms this statistical significance, with $p < 0.001$ for all configurations (Table 10). MRHDABC consistently achieves the lowest mean rank. The rank differences between MRHDABC and ACO-RNK, HTSMOGA, and AMOEA exceed the critical difference threshold across all scales, confirming statistical significance (Fig. 11). Box plot analysis (Figs. 12–13) demonstrates MRHDABC's superiority in both solution quality (lowest median) and algorithmic robustness (minimal variance).

Experimental results demonstrate MRHDABC's superiority for MRHPRTAS problem instances. This superiority is attributed to its innovative integration of heterogeneous robot handling and systematic load balancing mechanisms. These two mechanisms collectively address

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the fundamental challenges of multi-region task allocation.

Table 9. Wilcoxon signed-rank test results.

n-m	N	MRHDABC vs MODABC	MRHDABC vs ACO-RNK	MRHDABC vs HTSMOGA	MRHDABC vs AMOEA
20-3	20	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$
40-5	20	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$
60-6	20	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$
80-9	20	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$
Overall	80	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$

Table 10. Friedman-test results for different tasks and robots.

Algorithms	Mean Rank			
	$n = 20/m = 3$	$n = 40/m = 5$	$n = 60/m = 6$	$n = 80/m = 9$
MRHDABC	1.05	1.00	1.00	1.00
MODABC	2.10	2.00	2.00	2.50
ACO-RNK	2.85	3.30	3.70	4.00
AMOEA	4.70	3.70	3.30	2.50
HTSMOGA	4.30	5.00	5.00	5.00
CN	20.00	20.00	20.00	20.00
p-value	< 0.001	< 0.001	< 0.001	< 0.001
Crit. Diff $\alpha = 0.05$	1.36	1.36	1.36	1.36
Crit. Diff $\alpha = 0.1$	1.23	1.23	1.23	1.23

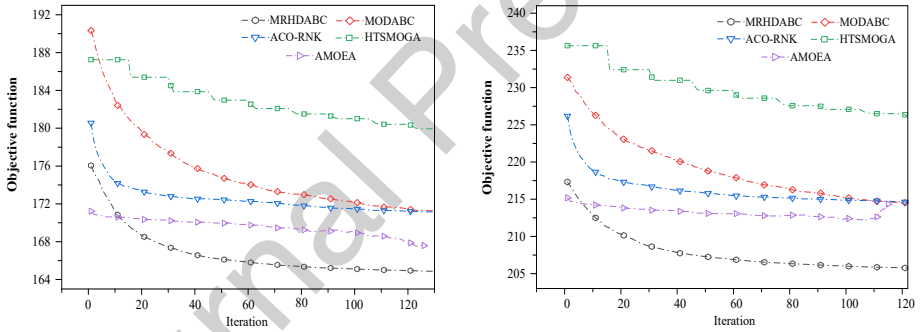


Fig. 10. Convergence analysis results under different task scales.

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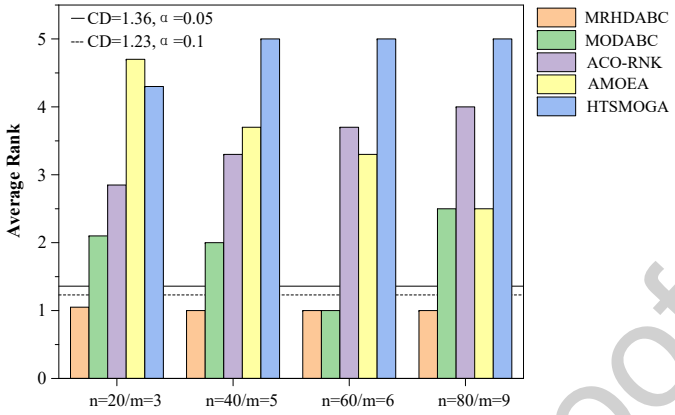


Fig. 11. Friedman-test results at different tasks and robots.

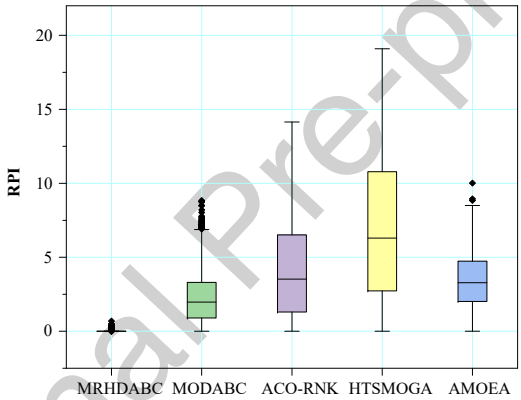


Fig. 12. The box plot for comparison algorithms.

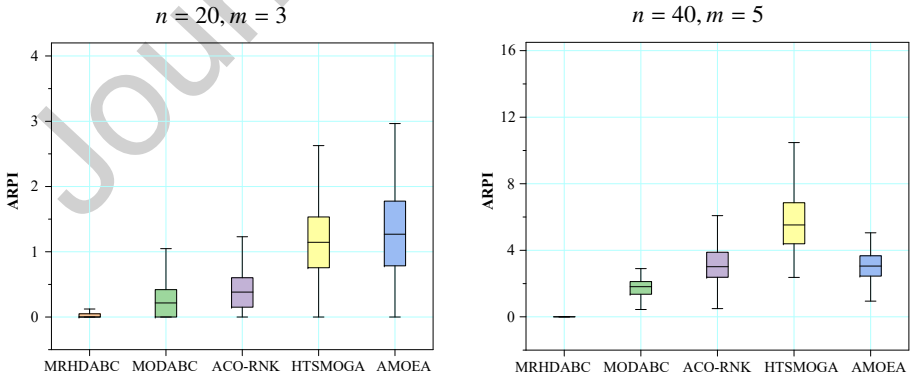


Fig. 13. The box plots of different numbers of tasks and robots.

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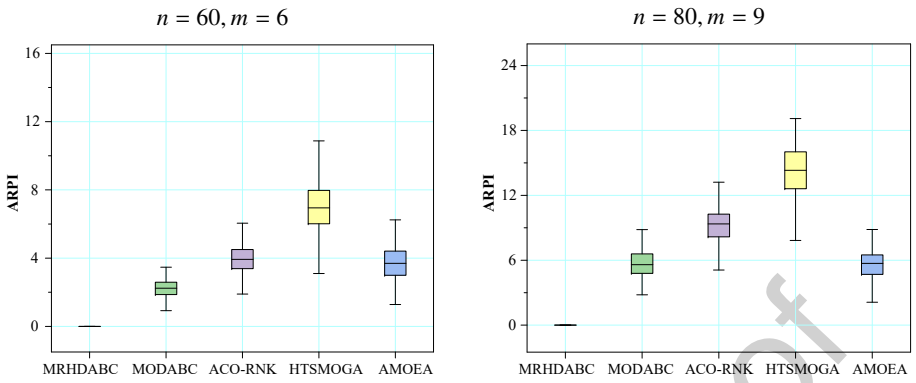


Fig. 13. (continued)

6. Conclusion

This paper addresses the MRHPRTAS problem, focusing on picking tasks executed by multiple heterogeneous picking robots across different regions in smart farms. The research objective is to minimize the Makespan. To solve this problem, three innovative heuristics are proposed: MRNEH-based, MRNEHIG-based, and IG-based heuristics, which generate high-quality initial solutions for MRHDABC. Another significant contribution of this research is the introduction of the critical picking robot-oriented optimization strategy and the systematic load balancing mechanism in MRHDABC. The critical picking robot-oriented optimization strategy effectively identifies system bottlenecks and achieves targeted optimization. Through designing single-robot, multi-robot, and critical-robot-oriented local search operators, and employing DOE and ANOVA techniques to tune algorithm parameters, the algorithm's search efficiency is further enhanced.

Comparative experiments confirm that MRHDABC significantly outperforms existing state-of-the-art methods, validating its applicability to the MRHPRTAS problem. Future research will consider more complex and practical constraints in the MRHPRTAS problem, such as multi-objective optimization incorporating energy consumption, cost, and stability, as well as the design of more effective heuristic methods based on problem knowledge. In addition, this study assumes that all picking robots have identical maximum load capacities, in order to focus on the modeled regional operation permissions and picking time coefficients. Future research can differentiate the maximum load capacity by robot type so that the model is applicable to scenarios involving heterogeneous picking robots with differing load capacities. This study addresses the problem in which the weight to be picked at each task point is known. Future research will investigate real-time replanning when the actual picked weight differs from the weight used in scheduling. Meanwhile, further research should investigate collaborative task allocation problems between picking robots and other agricultural robots, as well as robust optimization strategies addressing uncertainty factors such as weather conditions and equipment failures. This will expand the practical applications of related algorithms in smart agricultural systems.

CRediT authorship contribution statement

Ying Bi: Conceptualization, Methodology, Software, Validation, Writing - original draft. **Hongyan Sang:** Conceptualization, Methodology, Supervision, Writing – review & editing, Funding acquisition. **Quan-Ke Pan:** Resources, Supervision. **Hengwei Guo:** Supervision, Writing – review & editing. **Biao Zhang:** Resources, Supervision.

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Declaration of Generative AI and AI-assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used Claude (Anthropic) in order to improve the English language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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